GlucoTwin AI 2.0 · System Architecture

Synthetic-data patient twin · experimental 120-minute estimate · not for clinical use

ONLINE DATA, PREDICTION, AND PRESENTATION FLOW

Data streams
- Historical EHR
- Wearable / IoT
- Synthetic sources

Validation
- Pydantic checks
- Units + ranges
- Timezone / duplicate

Patient twin
- Profile + history
- Latest observations
- Updateable state

Temporal features
- Past-only lags
- Rolling summaries
- Training medians

ML prediction
- Saved model pipeline
- 120-minute event
- Uncalibrated score

Explainability
- Observed signals
- Global feature ranks
- Quality warnings

FastAPI
- Validated JSON routes
- Health + /docs
- Exact-origin CORS

SQLite
- Patient + readings
- Prediction history
- Local persistence

Dashboard
- Responsive UI
- Measured glucose trend
- Results + safety notes

OFFLINE TRAINING / EVALUATION · EXPLICIT COMMAND

Synthetic generator
- 36 profiles · 14 days
- 15-minute rows
- Fixed seed + marker

Label + features
- Past-only inputs
- Next 8-row event label
- Incomplete windows removed

Patient split
- Train / validation / test
- No person crosses groups
- Fixed split seed

Model comparison
- Prior / logistic baseline
- Random Forest candidate
- Select on validation PR-AUC

Saved artifacts
- joblib inference bundle
- Test metrics + confusion
- Synthetic limitations report

The generated synthetic-trained pipeline is loaded by the prediction route; training never implies clinical validity.

PROTOTYPE SAFEGUARDS

Synthetic-labelled records · parameterized SQL · exact-origin CORS · environment configuration

No authentication, production controls, clinical validation, regulatory clearance, or care use